The maximum size of a blackout that keeps every lattice rectangle identifiable

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Abstract

Black out some points of an $h \times m$ grid. The blackout is valid if every rectangle with corners in the grid, tilted or not, is determined by its white corners. Kagey asked for the largest size $f(h, m)$ of a valid blackout. For axis-aligned rectangles alone this is a Turán problem for bipartite graphs without cycles of length 4 and 6, and a bound of Neuwirth gives $f(h, m) \leqslant m + \lfloor h^2/4 \rfloor$ for $m \geqslant 3$. We prove that equality holds for every height $h \geqslant 3$ and every width $m \geqslant h^4/16 + O(h^3)$, by an explicit construction. Let m_h be the least width from which equality always holds. We prove that $\lfloor h^2/4 \rfloor \leqslant m_h \leqslant h^{2+o(1)}$. The upper bound places the columns that contain two black points at random and uses the local lemma. We also determine $f(h, m)$ for every $h \leqslant 6$ and every m, and for $h = 7$ and $m \geqslant 24$. Some of these small values rest on computer checks with verified certificates. On square grids the deletion method gives $f(n, n) \geqslant n^{6/5-o(1)}$, while the girth condition gives $f(n, n) \leqslant (1 + o(1))\, n^{4/3}$.

Mathematics Subject Classifications: 05C35, 05D40, 52C05, 68V05

1 Introduction

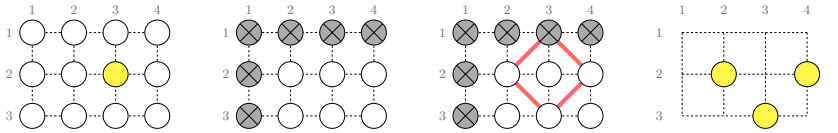
Start with an $h \times m$ grid of points and black out some of them. A game master then chooses a rectangle with corners in the grid, possibly a tilted one, and tells us which of its corners are white. The blackout is *valid* if this always determines the rectangle (Definition 4). Kagey [15, 14] asks for the largest size $f(h, m)$ of a valid blackout. Figure 1 shows one round. The known values of $f(h, m)$ are sequence A397315 in the OEIS [22].

Throughout the paper we put $q_h = \lfloor h^2/4 \rfloor$ and $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$. A tilted rectangle on h rows meets at most $W_h + 1$ consecutive columns (Lemma 9).

We first look at the axis-aligned rectangles alone. Record a blackout S by its *row-column graph* G_S, the bipartite graph that joins row j to column i when the point in column i and row j is black. For $m \geqslant 3$, the blackout distinguishes all axis-aligned

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(a) the point $(3,2)$ (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on a largest valid blackout of the 3×4 grid. Points are (column, row). Gray crossed circles are black. Panel (c) shows the red rectangle, and (d) its white corners.

rectangles exactly when G_S has no cycle of length 4 or 6 (Lemma 14). Hence $|S|$ is at most the largest number of edges of a bipartite graph of girth at least 8 with parts of sizes h and m. Neuwirth [21, Theorem 1(ii) and Proposition 17] proved that this number is at most $m + q_h$ when $m \geqslant q_h$, and that it is attained by the following graph. Take a complete bipartite graph on the h rows with parts as equal as possible. Subdivide each of its q_h edges by a column, and join every remaining column to one row. In the grid, this means that q_h columns contain 2 black points, on q_h different row pairs, and that every other column contains 1 black point. We call such columns *doubles* and *singletons*.

The graph G_S does not see the order of the columns, but the tilted rectangles do. Hence the problem for all rectangles is a problem about embeddings. We must order the columns of a subdivided triangle-free graph with pendant leaves, and choose the rows of the leaves, so that no 2 rectangles of the grid show the same white corners. Our first theorem says that this is possible at every height once the grid is wide enough.

Theorem 1 (large widths)**.** *Let $h \geqslant 3$ and put $M_h = q_h + (q_h - 1)W_h$. Then $f(h,m) = m + q_h$ for every $m \geqslant M_h$.*

The upper bound holds for every $m \geqslant 3$ (Theorem 15). For the lower bound we place the doubles $W_h + 1$ columns apart. The singletons are black in row 1 or in row h, and they switch between these 2 rows at every double (Figure 3). Note that $M_h = h^4/16 + O(h^3)$.

For $h \geqslant 3$ let m_h be the least integer $M \geqslant 1$ such that $f(h,m) = m + q_h$ for every $m \geqslant M$. We do not know whether $f(h,m+1) \geqslant f(h,m)+1$ always holds. Hence equality at one width does not obviously give equality at the next width, and this is why we define m_h through all larger widths. Theorem 1 gives $m_h \leqslant M_h$. Our second theorem finds the order of m_h up to a factor $h^{o(1)}$. Write $\tau(n)$ for the number of divisors of n, and put $\tau^*(h) = \max\{\tau(n) : 1 \leqslant n \leqslant h^2\}$.

Theorem 2 (the threshold)**.**

(a) For every $h \geqslant 4$ we have $m_h \geqslant q_h$.

(b) There is an absolute constant C such that $m_h \leqslant C\, h^2\, \tau^*(h)^2$ for every $h \geqslant 3$.

In particular, for every $\varepsilon > 0$ there is a constant C_ε with $q_h \leqslant m_h \leqslant C_\varepsilon h^{2+\varepsilon}$ for every $h \geqslant 4$.

Part (a) holds because equality needs q_h doubles (Corollary 19). For part (b) we use the odd and the even rows as the 2 parts, and we let every singleton be black in one of 2 fixed rows. We then choose the columns of the doubles independently at random in a window of length about $q_h\,\tau^*(h)^2$. A confusion between 2 rectangles needs a few black points in a special position, and for fixed rows the number of such positions is bounded by a divisor function. The local lemma gives a choice of columns with no confusion that involves 2 or more doubles. The confusions that involve at most 1 double are removed by fixing the singletons in a few columns near each double. The last statement of the theorem follows from the bound $\tau(n) \leqslant K_\delta n^\delta$ [11, Theorem 315].

Theorem 3 (small heights)**.** *The following hold. Parts* (c) *to* (e) *and the values* $f(4,4)$ *and* $f(4,5)$ *are computer-assisted.*

(a) $f(2,m) = m+1$ *and* $f(3,m) = m+2$ *for* $m \geqslant 3$.

(b) $f(4,4) = 7$, $f(4,5) = 8$, *and* $f(4,m) = m+4$ *for* $m \geqslant 6$.

(c) $f(5,m) = m+5$ *for* $5 \leqslant m \leqslant 7$, *and* $f(5,m) = m+6$ *for* $m \geqslant 8$.

(d) $f(6,m) = m+6$ *for* $m = 6, 7$, $f(6,m) = m+7$ *for* $8 \leqslant m \leqslant 11$, $f(6,m) = m+8$ *for* $12 \leqslant m \leqslant 14$, *and* $f(6,m) = m+9$ *for* $m \geqslant 15$.

(e) $f(7,7) = 14$, $f(7,8) = 16$, *and* $f(7,m) = m+12$ *for* $m \geqslant 24$.

Since $f(h,m) = f(m,h)$, $f(1,m) = m$, and $f(2,2) = 4$, this determines $f(h,m)$ for every $h \leqslant 6$ and every m. The value $f(3,m) = m+2$ was first proved in Lean by the Omega Institute [27]. Theorem 3 gives $m_4 = 6$, $m_5 = 8$, $m_6 = 15$, and $m_7 \leqslant 24$, while $q_h = 4, 6, 9, 12$ for these heights. For the infinite families in (c) to (e) we use the finite collar theorem (Theorem 35), which reduces the validity of a blackout padded by any number of boundary columns to one finite check. The upper bounds below $m + q_h$ rest on SAT instances with checked LRAT certificates (Section 7).

On square grids the 2 bounds are far apart. The deletion method gives $f(n,n) \geqslant c\,n^{6/5}(\log n)^{-2/5}$, while Neuwirth's bound for girth 8 gives $f(n,n) < n^{4/3} + \frac{2}{3}n$ (Section 8).

Related work. Choosing white points that distinguish all rectangles is a test cover problem [4, 16, 19]. If we only forbid fully black axis-aligned rectangles, we get the Zarankiewicz problem [8, 17, 25]. Füredi and Simonovits [9] survey such degenerate extremal problems. Bipartite graphs without short even cycles and with parts of unequal sizes are studied by de Caen and Székely [6, 7], Győri [10], Hoory [13], and Naor and Verstraëte [20]. With n vertices on each side, the largest graphs of girth 8 have $\Theta(n^{4/3})$ edges, and incidence graphs of generalized quadrangles attain this order [2]. As said above, the bound $m + q_h$ for the graph and its sharpness are due to Neuwirth [21]. In this part our contribution is Lemma 14, which translates rectangles into cycles, and the description of all cases of equality (Theorem 18). Everything about tilted rectangles is new. A related problem for tilted right angles in a grid is studied by Shkredov and Solymosi [26], who consider colorings and dense subsets of a grid without tilted isosceles right triangles. A

companion paper [23] counts the maximum blackouts at fixed height. We do not use its results.

2 Definitions and a criterion for validity

We use the following conventions.

(1) The $h \times m$ grid has h rows and m columns. The point (i, j) lies in column i and row j, and $(1, 1)$ is the top left point (Figure 1).

(2) A *rectangle* has 4 grid points as corners and positive area. We write $C(R)$ for the corner set of R. A rectangle is *axis-aligned* if its sides follow the grid lines, and *tilted* otherwise.

(3) A *blackout* is a set S of grid points, which we call black. The other points are white. The *presentation* of a rectangle R is $C(R) \setminus S$, the set of its white corners.

(4) Column i has the set $B_i = \{j : (i, j) \in S\}$ of *black rows*. It is *empty*, a *singleton*, or a *double* when $|B_i|$ is 0, 1, or 2.

Definition 4. A blackout S is *valid* if distinct rectangles have distinct presentations. We write $f(h, m)$ for the largest size of a valid blackout of the $h \times m$ grid.

Transposing the grid gives $f(h, m) = f(m, h)$. A grid with 1 row has no rectangles, so $f(1, m) = m$. The 2×2 grid has 1 rectangle, so $f(2, 2) = 4$. A rectangle is allowed to show no corners, but at most 1 rectangle can do so in a valid blackout.

Lemma 5. *Two rectangles R_1 and R_2 have the same presentation if and only if the set $C(R_1) \triangle C(R_2)$ is contained in S. Hence every subset of a valid blackout is valid.*

Proof. The 2 presentations agree exactly when every point that lies in just one of the sets $C(R_1)$ and $C(R_2)$ belongs to S. □

Lemma 6. *Any 3 corners of a rectangle determine it. Hence distinct rectangles have at most 2 common corners, and if they have the same presentation, then each shows at most 2 corners.*

Proof. Three corners form a right triangle. Its right angle is at the corner x adjacent to the other 2 corners y and z, and the fourth corner is $y + z - x$. □

Lemma 7. *Let the rectangles $R_1 \neq R_2$ have exactly 2 common corners p and q. If pq is a side of both or a diagonal of both, then $C(R_1) \triangle C(R_2)$ is the corner set of a rectangle.*

Proof. Suppose first that pq is a side of both. Then the other corners are $p + u, q + u$ and $p + v, q + v$, where $u \neq v$ are nonzero and perpendicular to $q - p$. These 4 points form a rectangle with sides $q - p$ and $v - u$. Suppose now that pq is a diagonal of both, with midpoint o. Then the other corners are $o \pm u$ and $o \pm v$ with $|u| = |v|$ and $v \neq \pm u$. The segments from $o - u$ to $o + u$ and from $o - v$ to $o + v$ have the same midpoint and the same length. Hence they are the diagonals of a rectangle. □

2.1 Tilted rectangles

Lemma 8. *A tilted rectangle R has a unique corner $\ell = (x, r)$ in its smallest column. The sides from ℓ are $(a, -b)$ and (c, d) with positive integers a, b, c, d and $ac = bd$, and the corners are*

$$(x, r), \qquad (x + a,\, r - b), \qquad (x + c,\, r + d), \qquad (x + a + c,\, r - b + d). \tag{1}$$

The rows of R are $r - b$ to $r + d$. Each of the smallest row, the largest row, and the largest column of R contains exactly 1 corner.

Proof. No side is vertical. Two corners in the smallest column would then be opposite, and their midpoint would force the other 2 corners into that column as well. Thus ℓ is unique, and the 2 sides from ℓ point strictly to the right. Their dot product is 0, so one points up and the other down, which gives (1) and $ac = bd$. The last statement follows from (1). □

We call $a + c$ the *span* of R.

Lemma 9. *Every tilted rectangle on h rows has span $a + c \leqslant W_h$.*

Proof. The rectangle fits in h rows, so $b + d \leqslant h - 1$. Also $(a - 1)(c - 1) \geqslant 0$. Hence $a + c \leqslant ac + 1 = bd + 1 \leqslant W_h$, where we used that $bd \leqslant \lfloor (b + d)^2/4 \rfloor$ for integers. □

The bound is attained by the sides $(1, -b)$ and (bd, d) with $b+d = h-1$ and $|b-d| \leqslant 1$.

Lemma 10. *Let R be a tilted rectangle of the $h \times m$ grid.*

(F1) R has at most 1 corner in row 1 and at most 1 corner in row h.

(F2) If R has corners in row 1 and in row h, then they are opposite.

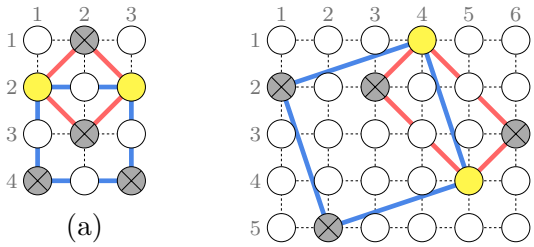
(F3) Two corners of R in one column are opposite, and so are 2 corners in one row.

Proof. (F1) follows from Lemma 8, since a corner in row 1 lies in the smallest row of R. (F3) holds because a vertical or a horizontal side would make R axis-aligned. For (F2), suppose that the 2 corners are adjacent. The other 2 corners are their translates by a nonzero vector $n = (n_1, n_2)$ perpendicular to the side. Staying in the grid requires $n_2 \geqslant 0$ at row 1 and $n_2 \leqslant 0$ at row h. Hence n is horizontal and the side is vertical, a contradiction. □

2.2 Mixed pairs

By Lemma 7, the remaining way for 2 rectangles to share 2 corners is the following.

Definition 11. A *mixed pair* is a pair (R_1, R_2) of rectangles with exactly 2 common corners p and q such that pq is a side of R_1 and a diagonal of R_2. Its *support* is $C(R_1) \triangle C(R_2)$. The support consists of the side AB of R_1 opposite to pq and of the diagonal CD of R_2 other than pq (Figure 2).



(a) (b)

Figure 2: Two mixed pairs. The common corners p, q are yellow, R_1 is blue, and R_2 is red. If the 4 points of the support are black, then both rectangles show exactly p and q.

In a mixed pair the rectangles are not both axis-aligned, since a side of an axis-aligned rectangle is horizontal or vertical and a diagonal is not. We record the equations of a support.

Lemma 12. *Let A, B, C, D be the support of a mixed pair as in Definition 11. Then there is a nonzero integer vector n such that*

$$n\cdot(B-A)=0, \qquad C+D=A+B+2n, \qquad |D-C|=|B-A|, \qquad D-C\neq\pm(B-A).$$

Proof. Name the common corners so that $p = A + n$ and $q = B + n$. Then n is a nonzero integer vector perpendicular to the side AB. The diagonals pq and CD of R_2 have the same midpoint and the same length, which gives the next 2 equations. If $D - C = \pm(B - A) = \pm(q - p)$, then $\{C, D\} = \{p, q\}$. □

Lemma 13 (criterion)**.** *Let S be a blackout of the $h \times m$ grid. Suppose that*

(i) the graph G_S has no cycle of length 4 or 6,

(ii) no tilted rectangle has 3 or more black corners, and

(iii) no mixed pair has all 4 points of its support black.

Then S is valid.

Proof. Suppose that $R_1 \neq R_2$ have the same presentation. By Lemma 5, the set $D = C(R_1)\triangle C(R_2)$ is black. By (i) and Lemma 14 below, whose proof does not use this lemma, the rectangles are not both axis-aligned. By Lemma 6 they have at most 2 common corners. With no common corner, both rectangles are fully black. With 1 common corner, both have 3 black corners. One of them is tilted, so both cases contradict (ii). Suppose finally that they have 2 common corners p and q. If pq is a side of both or a diagonal of both, then D is the corner set of a rectangle R_3 by Lemma 7, and R_3 is fully black. If R_3 is tilted, this contradicts (ii). If R_3 is axis-aligned, its corners give a cycle of length 4 in G_S, which contradicts (i). Otherwise (R_1, R_2) or (R_2, R_1) is a mixed pair with support D, which contradicts (iii). □

3 The row bound and the equality structure

A blackout S is *axis-valid* if distinct axis-aligned rectangles have distinct presentations. Recall that the row-column graph G_S joins row r_j to column c_i for each black point (i, j).

Lemma 14. *If G_S has no cycle of length 4 or 6, then S is axis-valid. The converse holds for $m \geqslant 3$.*

Proof. Let $R_1 \neq R_2$ be axis-aligned with the same presentation, so that $C(R_1) \triangle C(R_2) \subseteq S$. They cannot share opposite corners. Hence their common corners lie in one row or in one column, and by symmetry we can assume that they lie in one row a. With no common corner, R_1 is fully black, which gives a cycle of length 4. With 2 common corners (x, a) and (y, a), the other rows $b \neq c$ of R_1 and R_2 give the cycle $r_b\, c_x\, r_c\, c_y$. With just 1 common corner (x, a), write $R_1 = \{x, y\} \times \{a, b\}$ and $R_2 = \{x, z\} \times \{a, c\}$. Then $y \neq z$ and $b \neq c$, and the 6 black points give the cycle $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$.

Conversely, let $m \geqslant 3$. A cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ gives the rectangles $\{i, w\} \times \{j, j'\}$ and $\{i', w\} \times \{j, j'\}$ for a third column w, and both show exactly their white corners in column w. A cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ gives the rectangles $\{x, y\} \times \{a, b\}$ and $\{y, z\} \times \{a, c\}$. Both are black except possibly at (y, a). □

Theorem 15 (row bound)**.** *Let $m \geqslant 3$. Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points. Hence $f(h, m) \leqslant m + q_h$.*

In graph terms this is the bound of Neuwirth [21, Theorem 1(ii) and Proposition 17] for bipartite graphs without cycles of length 4 and 6, which he states for $m \geqslant q_h$. He contracts each column to the set of all pairs of its black rows and applies Mantel's theorem [18]. We give the short proof because we need its cases of equality. We join the black rows B_i of each nonempty column by a tree T_i instead of a complete graph, since a column with 3 black rows would give a triangle. Let H be the union of these trees, a graph on the h rows.

Lemma 16. *Let $m \geqslant 3$ and let S be axis-valid. For any choice of the trees, H is simple and triangle-free, and $e(H) = |S| - t$, where t is the number of nonempty columns.*

Proof. By Lemma 14, G_S has no cycle of length 4 or 6. A repeated edge of H would put the same row pair in 2 columns, which gives a cycle of length 4. Hence $e(H) = \sum_i (|B_i| - 1) = |S| - t$. Suppose that H contains a triangle $r_a r_b r_c$. Its edges do not all belong to one tree. If 2 columns i, i' supply them, then 2 of a, b, c belong to $B_i \cap B_{i'}$, which again gives a cycle of length 4. If 3 columns supply them, we get the cycle $r_a\, c_i\, r_b\, c_{i'}\, r_c\, c_{i''}$ of length 6. □

Lemma 17 (Mantel)**.** *Let H be a simple triangle-free graph on h vertices. Let a vertex of maximum degree Δ have neighbor set Γ, and let Ω be the set of the other $h - \Delta$ vertices, which spans $e(\Omega)$ edges. Then*

$$e(H) = \sum_{b \in \Omega} d(b) - e(\Omega) \leqslant (h - \Delta)\Delta - e(\Omega) \leqslant q_h.$$

Proof. The set Γ is independent, so every edge meets Ω. The degree sum over Ω counts the edges inside Ω twice. Then use $d(b) \leqslant \Delta$ and $(h-\Delta)\Delta \leqslant q_h$. □

Proof of Theorem 15. Lemmas 16 and 17 give $|S| = t + e(H) \leqslant m + q_h$. □

Theorem 18 (equality structure)**.** *Let $h \geqslant 4$ and $m \geqslant 3$. A blackout of the $h \times m$ grid with $m + q_h$ points is axis-valid if and only if*

(a) every column is a singleton or a double,

(b) the q_h doubles have distinct row pairs, and

(c) these pairs are the edges of a complete bipartite graph on the rows with parts of sizes $\lfloor h/2 \rfloor$ and $\lceil h/2 \rceil$.

Proof. Let S be axis-valid with $m + q_h$ points. Equality in the proof of the row bound forces $t = m$, $e(\Omega) = 0$, $(h-\Delta)\Delta = q_h$, and $d(b) = \Delta$ for every $b \in \Omega$. Thus H is the complete bipartite graph between Γ and Ω, as in (c), for every choice of the trees. Suppose that $|B_i| \geqslant 3$. Then T_i has a vertex v with 2 neighbors u and w, which lie in one part. Since $h \geqslant 4$, the part of v contains another vertex z. Replace the edge vw of T_i by uw. This gives another tree on B_i, and the new graph H contains uz, wz, and uw, which contradicts Lemma 16. This proves (a). The tree edges are now the row pairs of the doubles, which gives (b) and (c).

Conversely, assume (a) to (c). By (a) and (b), no 2 columns share 2 black rows, so G_S has no cycle of length 4. A cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ needs doubles on $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$, which contradicts (c). Now apply Lemma 14. □

Hence there are $p_h\, (m)_{q_h}\, h^{m-q_h}$ axis-valid blackouts with $m + q_h$ points, where $(m)_q$ is the falling factorial and p_h is the number of bipartitions in (c). The graph G_S of such a blackout is the extremal graph of Neuwirth from the introduction.

Corollary 19. *Let $h \geqslant 4$ and $3 \leqslant m < q_h$. Then $f(h, m) \leqslant m + q_h - 1$. Hence $m_h \geqslant q_h$ for $h \geqslant 4$.*

Proof. A valid blackout with $m + q_h$ points would have q_h doubles in different columns by Theorem 18. For the last statement, note that $q_h - 1 \geqslant 3$. □

This proves Theorem 2(a). We end the section with a lower bound that holds on every grid.

Lemma 20 (boundary cross)**.** *The first row and the first column of the $h \times m$ grid form a valid blackout. Hence $f(h, m) \geqslant h + m - 1$.*

Proof. The graph G_S is a tree, so (i) of Lemma 13 holds. Let R be a tilted rectangle with corners (1). By Lemma 8, its only possible corner in column 1 is ℓ, and its only possible corner in row 1 is $\ell + (a, -b)$. Hence (ii) holds, and 2 black corners of a tilted rectangle are always adjacent. Let (R_1, R_2) be a mixed pair with black support A, B, C, D. The corners C, D of R_2 are black and opposite, so R_2 is axis-aligned and R_1 is tilted. Then

$\{A, B\} = \{\ell, \ell + (a, -b)\}$ with $\ell = (1, 1+b)$, and the common corners $p = (1+c, 1+b+d)$ and $q = (1 + a + c, 1 + d)$ lie in columns and rows at least 2. The other corners C, D of the axis-aligned rectangle with diagonal pq also lie in columns and rows at least 2, so they are white, a contradiction. Thus (iii) holds. □

4 An explicit construction at every height

We now prove Theorem 1. We place the doubles far apart and let the singletons between them be black in a boundary row. Call a singleton a *sea* if its black point lies in row 1 or in row h.

Proposition 21 (alternating seas)**.** *Let $h \geqslant 3$ and let H be a triangle-free graph on the rows with edges $e_1, \dots, e_q$, where $q \geqslant 1$. Let Z be the blackout of the $h \times m$ grid with the following columns. The columns $p_1 < \cdots < p_q$ are doubles, where column p_j is black exactly in the 2 rows of e_j and $p_{j+1} - p_j \geqslant W_h + 1$. Every other column is a sea. The seas to the left of p_1 are black in row 1, and the row of the seas switches between 1 and h at every double. Then Z is valid and $|Z| = m + q$.*

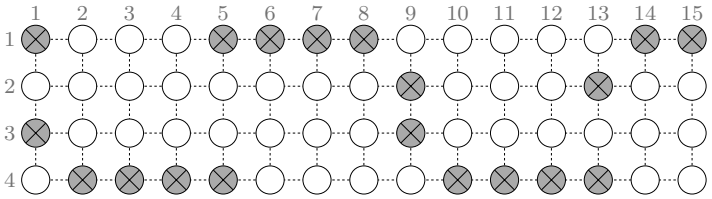


Figure 3: The blackout of Proposition 21 for $h = 4$ and $m = 15$, with the 4 edges of the complete bipartite graph between $\{1, 2\}$ and $\{3, 4\}$. Here $W_4 + 1 = 4$.

Proof of Theorem 1. Theorem 15 gives the upper bound. Take H to be the complete bipartite graph with parts of sizes $\lfloor h/2 \rfloor$ and $\lceil h/2 \rceil$, so $q = q_h$, and place the doubles at $p_j = 1 + (j-1)(W_h + 1)$. The last double is in column $1 + (q_h - 1)(W_h + 1) = M_h$. Hence Proposition 21 gives a valid blackout with $m + q_h$ points for every $m \geqslant M_h$. □

Proof of Proposition 21. Each double adds 1 point to a column, so $|Z| = m + q$. We check the 3 conditions of Lemma 13.

Condition (i). Only doubles have 2 black rows, and their pairs are distinct, so G_Z has no cycle of length 4. A cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ would need doubles on $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$, a triangle in H.

The black corners of a tilted rectangle. Let R be a tilted rectangle. In this proof, *the columns of R* are all columns from its smallest to its largest column, not only the columns of its corners. The span of R is at most W_h (Lemma 9), so its columns contain at most 1 double. With no double, all black points in the columns of R lie in one boundary row, and R has at most 1 black corner by (F1). Otherwise, shift the double to column 0 and

write its black rows as $a < b$. Reflecting the rows if needed, we can assume that the seas in the columns of R are black in row 1 to the left of column 0 and in row h to the right.

Let K be the set of black corners of R. By (F1), the elements of K can only be one point $u = (x_1, 1)$ with $x_1 < 0$, one point $v = (x_2, h)$ with $x_2 > 0$, and points $(0, c)$ with $c \in \{a, b\}$. A corner $(0, 1)$ excludes u, and a corner $(0, h)$ excludes v. We now list the sets K with $|K| \geqslant 2$. In each case we find the other corners of R.

(a) $K = \{u, v\}$. These corners are opposite by (F2). The other diagonal has midpoint $\left(\frac{x_1+x_2}{2}, \frac{1+h}{2}\right)$ and endpoints at this midpoint plus and minus a vector w with $|w|^2 = \frac{1}{4}\big((x_2 - x_1)^2 + (h-1)^2\big)$. By (F1) the endpoints lie in rows 2 to $h-1$. Hence $|w_2| \leqslant \frac{h-3}{2}$ and $w_1^2 > \frac{1}{4}(x_2 - x_1)^2$. So one endpoint lies to the left of column x_1 and the other to the right of column x_2.

(b) $K = \{(0, a), (0, b)\}$. These corners are opposite by (F3). The other diagonal has midpoint $\left(0, \frac{a+b}{2}\right)$ and length $b - a$, and it is not vertical. Hence its endpoints lie in columns $-w_1$ and w_1 for some $w_1 \neq 0$, and in rows strictly between a and b.

(c) $K = \{u, d\}$ with $d = (0, c)$, and u, d are opposite. Here $c \geqslant 2$ by (F1). The other diagonal has midpoint $\left(\frac{x_1}{2}, \frac{1+c}{2}\right)$ and endpoints at this midpoint plus and minus a vector w with $|w|^2 = \frac{1}{4}\big(x_1^2 + (c-1)^2\big)$. The endpoints are not in row 1 by (F1). They are not in row h by (F2), since the corner opposite to u is d. They are not in column 0 by (F3), for the same reason. Since their rows are at least 2, we get $|w_2| \leqslant \frac{c-3}{2}$. In particular $c \geqslant 3$, and $w_1^2 > \frac{1}{4}x_1^2$. Thus one endpoint lies to the right of column 0 and the other to the left of column x_1.

(d) $K = \{u, d\}$ with $d = (0, c)$, and u, d are adjacent. Again $c \geqslant 2$. The side $d - u$ is $(|x_1|, c-1)$, and the other corners are $u + n$ and $d + n$ with $n = t\,(-(c-1), |x_1|)$ for some real $t \neq 0$. The corner $u + n$ lies in row $1 + t|x_1|$, so $t > 0$. Hence both corners lie in columns to the left of column 0 and in rows at least 2.

(e) $K = \{v, d\}$ with $d = (0, c)$. The half turn $(x, j) \mapsto (-x, h + 1 - j)$ exchanges the 2 seas and fixes column 0. It reduces this case to (c) or (d).

(f) $|K| \geqslant 3$. This is impossible. If K contains $(0, a)$ and $(0, b)$, then the other 2 corners lie in rows strictly between a and b by (b), so they are neither u nor v, and they are not in column 0. Otherwise $K = \{u, d, v\}$ with $d = (0, c)$. Then u and v are opposite by (F2) and d is adjacent to both. So $(u - d) \cdot (v - d) = 0$, that is $x_1 x_2 = (c-1)(h-c)$. The left side is negative and the right side is not.

In (a) to (d), the 2 other corners lie in columns different from 0 and in rows 2 to $h-1$, or to the left of column 0 and in rows at least 2. The seas in these columns are black only in row 1 or only in row h, so the 2 other corners are white. By (e) the same holds in the reflected cases. We have proved the following 3 statements for a tilted rectangle R.

(1) R has at most 2 black corners. Hence condition (ii) holds.

(2) If R has 2 opposite black corners, then its 2 white corners lie in rows 2 to $h-1$, one of them to the left and the other to the right of the double in the columns of R.

(3) If R has 2 adjacent black corners, then both white corners of R lie on one side of the double in the columns of R. On this side the seas in the columns of R are black in one boundary row, and the white corners are not in that row.

Condition (iii). Let (R_1, R_2) be a mixed pair with common corners p, q whose support is black. A tilted rectangle of the pair has 2 black corners in the support, so its corners p, q are white by (1).

Suppose that R_1 is axis-aligned. Then R_2 is tilted with 2 opposite black corners. By (2), p and q lie in different columns $x < z$ on the 2 sides of the double of R_2. So the side pq of R_1 is horizontal. The other 2 corners of R_1 are black points of the columns x and z in a common row. But these columns are seas, one black only in row 1 and the other only in row h, a contradiction.

Suppose that R_2 is axis-aligned. Then R_1 is tilted with 2 adjacent black corners. By (3), $p = (x, j)$ and $q = (z, k)$ lie in sea columns that are black only in one boundary row σ, and $j, k \neq \sigma$. The other corners of R_2 are (x, k) and (z, j), which are white, a contradiction.

Suppose finally that both are tilted. The side opposite to pq in R_1 is black, so (3) applies to R_1. Let D_1 be the column of the double in the columns of R_1. Then p and q lie on one side of column D_1. The diagonal of R_2 other than pq is black, so (2) applies to R_2. Let D_2 be the column of the double in the columns of R_2. Then D_2 lies strictly between the columns of p and q. Hence $D_2 \neq D_1$, and D_2 is a second double in the columns of R_1, a contradiction. Thus (iii) holds, and Z is valid by Lemma 13. □

Remark 22. The proof uses the spacing $W_h + 1$ only to keep 2 doubles out of the columns of one tilted rectangle. The spacing W_h is not enough. For $h = 4$ and $h = 6$, the blackout with spacing W_h is not valid for the complete bipartite graph between the rows $1, \dots, h/2$ and the other rows, with the doubles in lexicographic order of their row pairs. The threshold M_h is far from m_h for small heights. For example $M_5 = 31$, while $m_5 = 8$.

5 The threshold is nearly quadratic

In this section we prove Theorem 2(b). A *parity edge* is a pair $\{r, s\}$ of rows with r odd and s even. There are q_h parity edges, and they are the edges of a complete bipartite graph as in Theorem 18(c). Our blackouts have one double for each parity edge and a singleton in every other column. All singletons of a long block of columns are black in one of 2 fixed rows, which we call the *sea rows*.

5.1 Obstructions

It is convenient to work in the strip $\Sigma_h = \mathbb{Z} \times \{1, \dots, h\}$. A rectangle *of the strip* has its corners in Σ_h. The lemmas of Section 2 hold in the strip with the same proofs. A *configuration* is a subset of Σ_h, whose points we call black.

Definition 23. We use 3 kinds of *obstructions*.

(1) A *corner triple* consists of 3 corners of a tilted rectangle of the strip.

(2) A *mixed quadruple* is the support $\{A, B, C, D\}$ of a mixed pair of the strip whose side AB is not vertical.

(3) Let R be a tilted rectangle of the strip that has a vertical diagonal pq, where the rows of p and q have different parity. The other 2 corners of R form a *rival pair*.

A configuration *contains* an obstruction if all points of the obstruction are black.

The points of an obstruction always lie in $2W_h + 1$ consecutive columns. For corner triples and rival pairs this holds because a tilted rectangle has span at most W_h. In a mixed pair (R_1, R_2) with common corners p, q, at least one rectangle is tilted. If both are tilted, they share the corner p and have span at most W_h. If R_2 is axis-aligned, its columns are those of p and q, which are columns of R_1. If R_1 is axis-aligned and pq is not vertical, then pq is horizontal, and the columns of R_1 are again those of p and q. A mixed pair with a vertical side pq does not give a mixed quadruple. It gives a rival pair (Lemma 24).

Lemma 24 (reduction)**.** *Let S be a blackout of the $h \times m$ grid. Suppose that every column is a singleton or a double, that the doubles have distinct row pairs, and that these pairs are parity edges. If S contains no obstruction, then S is valid.*

Proof. We check the conditions of Lemma 13. The graph G_S has no cycle of length 4 because the doubles have distinct pairs. A cycle of length 6 would need 3 doubles whose pairs form a triangle, but the parity edges form a bipartite graph. A tilted rectangle with 3 black corners gives a corner triple. Let (R_1, R_2) be a mixed pair with common corners p, q and black support A, B, C, D. If the side AB is not vertical, the support is a mixed quadruple. Suppose that AB is vertical. Then A and B are 2 black points of one column, so their rows form a parity edge. The side pq of R_1 is vertical with the same 2 rows. Now pq is a diagonal of R_2, so R_2 is tilted, and $\{C, D\}$ is a rival pair. □

The next 3 lemmas show that some simple configurations contain no obstructions.

Lemma 25 (rival pairs)**.** *Let $\{C, D\}$ be a rival pair with C in row c, D in row c', and $c \leqslant c'$. Then $c' - c$ is odd, and there is an integer $j \geqslant 1$ with*

$$c - j \geqslant 1, \qquad c' + j \leqslant h, \qquad 4j(c' - c + j) = (C_x - D_x)^2,$$

where C_x and D_x are the columns of C and D. In particular, no point of a rival pair lies in row 1 or in row h, and the 2 points lie in rows of different parity.

Proof. Let R and pq be as in Definition 23(3), and let L be the length of pq. Then L is an odd integer. The diagonals pq and CD have the same midpoint and the same length. Put $t = C_x - D_x$ and $d = c' - c$, so that $L^2 = t^2 + d^2$. Here $t \neq 0$, since CD is not vertical, and so $L > d$. The rows of p and q are $\frac{1}{2}(c + c' - L)$ and $\frac{1}{2}(c + c' + L)$. They are integers, so $L - d$ is even. Write $L = d + 2j$ with $j \geqslant 1$. Then d is odd, the rows of p and q are $c - j$ and $c' + j$, and $t^2 = L^2 - d^2 = 4j(d + j)$. □

Lemma 26 (two sea rows)**.** *Let g be an even integer with $2 \leqslant g \leqslant h$ and $2g-1 > h$. No obstruction has all its points in the rows 1 and g.*

Proof. Suppose that 3 corners of a tilted rectangle R lie in the rows 1 and g. Then 2 of them lie in one row y_0, and they are opposite by (F3). So no other corner lies in row y_0, and the third corner lies in the other row y_1. The center of R is in row y_0, so the fourth corner is in row $2y_0 - y_1$. This row is $2 - g < 1$ or $2g - 1 > h$, a contradiction.

Let $\{A, B, C, D\}$ be a mixed quadruple in the rows 1 and g, and let $n = (n_1, n_2)$ be as in Lemma 12. Write $B - A = (x, y)$, where $x \neq 0$. The equation $C + D = A + B + 2n$ shows that the rows of C and D have the same sum modulo 2 as the rows of A and B. Since $1 + g$ is odd, either both pairs lie in one row each, or both pairs meet both rows. In the first case $y = 0$, so $n_1 x = 0$ gives $n_1 = 0$, and $D - C$ is horizontal with $|D - C| = |x|$. Then $D - C = \pm(B - A)$, a contradiction. In the second case the row sums are equal, so $n_2 = 0$. Then $n_1 x = 0$ gives $n = 0$, a contradiction.

By Lemma 25, a rival pair has no point in row 1 and does not lie in one row. □

A *chain* is a set of points of $\mathbb{Z}^2$ in which any 2 points P, P' satisfy $P \leqslant P'$ or $P' \leqslant P$ in both coordinates.

Lemma 27 (chains)**.** *A chain contains no corner triple and no mixed quadruple.*

Proof. Let Q_0, Q_1, Q_2 be a corner triple with the right angle at Q_0. The vectors $Q_1 - Q_0$ and $Q_2 - Q_0$ are sides of a tilted rectangle, so all their coordinates are nonzero. In a chain, each of them has 2 positive or 2 negative coordinates. Then their dot product is not 0, a contradiction.

Let $\{A, B, C, D\}$ be a mixed quadruple in a chain, with n as in Lemma 12. We can name the points so that $A \leqslant B$ and $C \leqslant D$. Put $U = B - A$ and $\varphi(P) = (P - A) \cdot U$. Then $\varphi(A) = 0$ and $\varphi(B) = |U|^2$. The equations $C + D = A + B + 2n$ and $n \cdot U = 0$ give $\varphi(C) + \varphi(D) = |U|^2$. Also $|\varphi(D) - \varphi(C)| = |(D - C) \cdot U| < |U|^2$, because $|D - C| = |U|$ and $D - C \neq \pm U$. Hence $0 < \varphi(P) < |U|^2$ for $P \in \{C, D\}$. The coordinates of U are nonnegative. So a point $P \leqslant A$ has $\varphi(P) \leqslant 0$, and a point $P \geqslant B$ has $\varphi(P) \geqslant |U|^2$. Since the 4 points form a chain, we get $A \leqslant C \leqslant D \leqslant B$. Then $0 \leqslant D - C \leqslant B - A$ in both coordinates, and $|D - C| = |B - A|$ forces $D - C = B - A$, a contradiction. □

Finally we count obstructions with given rows. Recall that $\tau^*(h) = \max\{\tau(n) : 1 \leqslant n \leqslant h^2\}$.

Lemma 28 (counting)**.** *Fix rows y_0, y_1, y_2, y_3.*

(a) There are at most $2\tau^*(h)$ triples (Q_0, Q_1, Q_2) such that $\{Q_0, Q_1, Q_2\}$ is a corner triple with the right angle at Q_0, the point Q_i lies in row y_i, and Q_0 lies in column 0.

(b) There are at most $4\tau^*(h)$ quadruples (A, B, C, D) that form a mixed quadruple with side AB and diagonal CD, with these points in the rows y_0, y_1, y_2, y_3 and with A in column 0.

(c) There are at most $2\tau^*(h)$ rival pairs $\{C, D\}$ with $C = (0, y_0)$ and D in row y_1.

Proof. (a) Write $Q_1 - Q_0 = (x_1, y_1 - y_0)$ and $Q_2 - Q_0 = (x_2, y_2 - y_0)$. All 4 coordinates are nonzero, and $x_1 x_2 = -(y_1 - y_0)(y_2 - y_0)$. So x_1 is a positive or negative divisor of a fixed integer k with $1 \leqslant |k| \leqslant h^2$, and x_1 determines x_2.

(b) Let $n = (n_1, n_2)$ be as in Lemma 12, and write $B - A = (x, y)$ and $D - C = (z, v)$. The rows fix y, v, and n_2, and we have $x \neq 0$, $n_1 x = -n_2 y$, and $x^2 + y^2 = z^2 + v^2$. Suppose first that $y \neq 0$. Then $n_2 \neq 0$, since otherwise $n = 0$. So x is a divisor of the integer $n_2 y$, where $1 \leqslant |n_2 y| \leqslant h^2$. There are at most $2\tau^*(h)$ such x. Each of them determines n_1 and allows at most 2 values of z. Suppose now that $y = 0$. Then $n_1 = 0$ and $x^2 - z^2 = v^2$. Here $v \neq 0$, since otherwise $D - C = \pm(B - A)$. Hence $|x| - |z|$ is a divisor of v^2, and it determines $|x|$ and $|z|$. This gives at most $4\tau(v^2) \leqslant 4\tau^*(h)$ pairs (x, z). In both cases x, z, and n determine $B = A + (x, y)$ and then C and D, by $C + D = A + B + 2n$.

(c) By Lemma 25, the column t of D satisfies $t^2 + d^2 = L^2$ with $d = |y_1 - y_0| \geqslant 1$ and an integer L. Then $L - |t|$ is a divisor of d^2, and it determines $|t|$. □

5.2 One double

Fix an even integer g with $2 \leqslant g \leqslant h$ and $2g - 1 > h$. The sea rows are 1 and g.

Definition 29. A parity edge $e = \{r, s\}$ with r odd is *good for g* if $r \leqslant g$, $s \leqslant g$, and $j(g - r + j)$ is not a perfect square for any integer j with $1 \leqslant j \leqslant \min(r - 1, h - g)$. Its *gadget* is the configuration

$$G_e = \{(x, 1) : x < 0\} \cup \{(0, r), (0, s)\} \cup \{(x, g) : x > 0\}.$$

An obstruction is *private to e* if it has a point in column 0, all its points in column 0 lie in the rows r and s, and all its other points lie in the rows 1 and g. Let A_e be the set of integers $a \neq 0$ such that some corner triple or mixed quadruple private to e has a point in column a.

The set A_e lists the columns near a double on e whose singletons can take part in a confusion with this double alone. We will fix the singletons in these columns as in the gadget.

Lemma 30 (one double)**.** *Let e be good for g.*

(a) We have $|A_e| \leqslant 2^{14}\,\tau^*(h)$, and $|a| \leqslant 2W_h$ for every $a \in A_e$.

(b) The gadget G_e contains no corner triple and no mixed quadruple.

(c) No rival pair consists of a point $(0, c)$ with $c \in e$ and a point in row 1 or g.

Proof. (a) In a private obstruction every point lies in one of the 4 rows $r, s, 1, g$. By Lemma 28(a), there are at most $4^3 \cdot 2\tau^*(h)$ corner triples with rows in this set, up to horizontal translations. Each of them has at most 3 translates with a point in column 0, and each translate has at most 2 other columns. This gives at most $768\,\tau^*(h)$ elements of

A_e. In the same way, Lemma 28(b) gives at most $4^4 \cdot 4\tau^*(h) \cdot 4 \cdot 3 = 12288\,\tau^*(h)$ elements. The bound on $|a|$ holds because an obstruction lies in $2W_h + 1$ consecutive columns.

(b) Since $r, s \leqslant g$, the gadget is a chain. Apply Lemma 27.

(c) Let $\{(0, c), D\}$ be such a rival pair. By Lemma 25, D is not in row 1. So D is in row g, and c is odd, that is $c = r < g$. Lemma 25 gives an integer j with $1 \leqslant j \leqslant \min(r-1, h-g)$ such that $4j(g-r+j)$ is a perfect square. Then $j(g-r+j)$ is a perfect square, which contradicts Definition 29. □

5.3 Many doubles

Let $\mathcal{E}$ be a set of parity edges that are good for g. We give each $e \in \mathcal{E}$ a column X_e, and we put $A_e^* = A_e \cup \{0\}$. For a set $L \subseteq \mathcal{E}$ with $2 \leqslant |L| \leqslant 4$, we say that the columns $(X_e)_{e \in L}$ are *bad* if at least one of the following holds.

(B1) $L = \{e, f\}$ and the sets $X_e + A_e^*$ and $X_f + A_f^*$ intersect.

(B2) $L = \{e, f\}$ and $\{(X_e, c), (X_f, c')\}$ is a rival pair for some $c \in e$ and $c' \in f$.

(B3) The columns X_e with $e \in L$ are distinct, and there is a corner triple or a mixed quadruple Q with the following 2 properties. Every point of Q in a column X_e with $e \in L$ lies in a row of e, and every other point of Q lies in row 1 or g. For every $e \in L$, at least one point of Q lies in column X_e.

Lemma 31 (placement)**.** *There is an absolute constant c_0 with the following property. Let $\mathcal{E}$ be a set of $n \geqslant 1$ parity edges that are good for g, and let $N \geqslant c_0\, n\, \tau^*(h)^2$ be an integer. Then there are columns $X_e \in \{1, \dots, N\}$ for $e \in \mathcal{E}$ such that $(X_e)_{e \in L}$ is not bad for any $L \subseteq \mathcal{E}$ with $2 \leqslant |L| \leqslant 4$.*

Proof. Choose the X_e independently and uniformly from $\{1, \dots, N\}$, and let E_L be the event that $(X_e)_{e \in L}$ is bad. Write $\tau^* = \tau^*(h)$.

We first bound $\Pr(E_L)$. Fix one edge $e_0 \in L$. Each of the conditions below fixes the differences $X_e - X_{e_0}$ for all $e \in L$, so it holds with probability at most $N^{-(|L|-1)}$. For (B1) there are at most $(|A_e| + 1)(|A_f| + 1) \leqslant 2^{30}\tau^{*2}$ possible differences $X_f - X_e$, by Lemma 30(a). For (B2) there are 4 choices of (c, c'). By Lemma 28(c), the pair $\{(X_e, c), (X_f, c')\}$ is a rival pair only if $X_f - X_e$ is one of at most $2\tau^*$ integers.

For (B3), let Q be as in (B3). If Q is a corner triple, list its points as (Q_0, Q_1, Q_2) with the right angle at Q_0. If Q is a mixed quadruple, list them as (A, B, C, D) with the side AB and the diagonal CD. Give each point of the list a *type*. A point in a column X_e with $e \in L$ gets the type (e, c), where $c \in e$ is its row. This is well defined because the columns X_e are distinct. Every other point gets its row, which is 1 or g, as its type. There are at most 10 types, so there are at most $10^3 + 10^4$ lists of types, and every $e \in L$ occurs in the list of Q. Now fix a list of types in which every $e \in L$ occurs. It fixes the rows of the points. By Lemma 28(a) and (b), at most $4\tau^*$ lists of points Q^* of the same kind have these rows and their first point in column 0, and Q is a horizontal translate of one of them. Fix Q^* as well. If Q is the translate of Q^* by x columns, then X_e is x plus

the column of every point of Q^* of type (e, c). Hence Q^* and the list of types determine all differences $X_e - X_{e_0}$ with $e \in L$. Altogether we get $\Pr(E_L) \leqslant p_{|L|}$, where

$$p_2 = c_1\,\tau^{*2}N^{-1}, \qquad p_3 = c_1\,\tau^* N^{-2}, \qquad p_4 = c_1\,\tau^* N^{-3}, \qquad c_1 = 2^{31}.$$

The event E_L depends only on the variables X_e with $e \in L$. Hence E_L is mutually independent of the family of all events $E_{L'}$ with $L' \cap L = \emptyset$. We apply the local lemma in its general form [1, Lemma 5.1.1] with the weights $x_L = 2p_{|L|}$. Put $c_0 = 48c_1$. An edge e lies in at most n^{k-1} sets L of size k. For $N \geqslant c_0 n\tau^{*2}$ this gives

$$\sum_{L \ni e} x_L \leqslant 2c_1\Big(\frac{n\tau^{*2}}{N} + \frac{n^2\tau^*}{N^2} + \frac{n^3\tau^*}{N^3}\Big) \leqslant \frac{6c_1}{c_0} = \frac{1}{8}.$$

In particular $x_L < 1$. Every set L has at most 4 elements, so the weights of the sets L' that meet L sum to at most $\frac{1}{2}$. Thus

$$x_L \prod_{L' \cap L \neq \emptyset} (1 - x_{L'}) \geqslant x_L\Big(1 - \sum_{L' \cap L \neq \emptyset} x_{L'}\Big) \geqslant \frac{x_L}{2} = p_{|L|} \geqslant \Pr(E_L).$$

By the local lemma, with positive probability no event E_L occurs. □

We now show that such columns give a configuration without obstructions, whatever the other singletons are. We state this for a window of columns, since for odd h we will use several blocks with different sea rows.

Lemma 32 (one block)**.** *Let $\mathcal{E}$ be a set of parity edges that are good for g, and let $(X_e)_{e\in\mathcal{E}}$ be integers such that $(X_e)_{e\in L}$ is not bad for any $L \subseteq \mathcal{E}$ with $2 \leqslant |L| \leqslant 4$. Let $I \subseteq \mathbb{Z}$ be an interval and let B be a configuration with the following 3 properties for every column $x \in I$.*

(1) If $x = X_e$ for some $e \in \mathcal{E}$, then column x is black exactly in the 2 rows of e.

(2) If $x = X_e + a$ for some $e \in \mathcal{E}$ and $a \in A_e$, then column x is black exactly in row 1 if $a < 0$, and exactly in row g if $a > 0$.

(3) Otherwise column x is black in exactly 1 point, which lies in row 1 or g.

Then B contains no obstruction whose points all lie in columns of I.

Since (B1) fails for all pairs, the columns X_e are distinct, and (1) and (2) never apply to the same column or give 2 different rules for it. Note that the conditions (B1) to (B3) only depend on the differences of the columns, so they are not affected by a translation of all columns.

Proof. Let Q be an obstruction in the columns of I whose points are black. Let L be the set of edges $e \in \mathcal{E}$ such that column X_e contains a point of Q. By (1), a point of Q in a column X_e with $e \in L$ lies in a row of e. Every other point of Q lies in a column that is

not X_f for any $f \in \mathcal{E}$, since otherwise $f \in L$. By (2) and (3), it is the only black point of its column and lies in row 1 or g.

Suppose first that Q is a rival pair. Its 2 points lie in different columns. If L is empty, this contradicts Lemma 26. If $L = \{e\}$, then Q consists of a point (X_e, c) with $c \in e$ and a point in row 1 or g, and its translate by $-X_e$ columns contradicts Lemma 30(c). If $L = \{e, f\}$, then $Q = \{(X_e, c), (X_f, c')\}$ with $c \in e$ and $c' \in f$, so (B2) holds for L.

Suppose now that Q is a corner triple or a mixed quadruple. If L is empty, this contradicts Lemma 26. Suppose that $|L| \geqslant 2$. Then $|L| \leqslant 4$, because Q has at most 4 points. The columns X_e with $e \in L$ are distinct, the first property in (B3) was shown at the start of the proof, and the second property is the definition of L. So (B3) holds for L, a contradiction. Suppose finally that $L = \{e\}$, and translate Q by $-X_e$ columns to an obstruction Q'. Then Q' is private to e. Hence every point of Q' outside column 0 lies in a column $a \in A_e$, and the corresponding point of Q lies in column $X_e + a$. This column is not a column X_f, so (2) applies to it, and by the paragraph before the proof (2) gives only one rule for it. So this point is in row 1 if $a < 0$ and in row g if $a > 0$. Thus Q' is contained in the gadget G_e, which contradicts Lemma 30(b). □

5.4 Proof of Theorem 2(b)

By Theorem 15 we only need valid blackouts with $m + q_h$ points. For $h \leqslant 8$ the claim follows from Theorem 1 after increasing C. So let $h \geqslant 9$.

Even h. Take $g = h$. Every parity edge $\{r, s\}$ is good for h, since $\min(r-1, h-g) = 0$. Let $\mathcal{E}$ be the set of all q_h parity edges and let $N = \lceil c_0 q_h \tau^*(h)^2 \rceil$. Take the columns X_e of Lemma 31, and let B be a configuration as in Lemma 32 with $I = \mathbb{Z}$. For example, let every column in (3) be black in row 1. For $m \geqslant N$ let S be the set of points of B in the columns 1 to m. It contains all doubles, so $|S| = m + q_h$. By Lemma 32, S contains no obstruction, and it is valid by Lemma 24. Hence $m_h \leqslant N$.

Odd h. Now $g = h$ is odd, so Lemma 26 does not apply to the sea rows 1 and h. We use the 4 pairs of sea rows

$$\{1, h-1\}, \qquad \{1, h-3\}, \qquad \{2, h\}, \qquad \{4, h\}.$$

The last 2 are the images of the first 2 under the reflection $\rho(x, j) = (x, h + 1 - j)$. For odd h this reflection maps parity edges to parity edges and obstructions to obstructions. Both $g = h - 1$ and $g = h - 3$ are even and satisfy $2g - 1 > h$, since $h \geqslant 9$.

We claim that for every parity edge $e = \{r, s\}$, with r odd, one of e and $\rho(e)$ is good for $h - 1$ or for $h - 3$. For $g = h - 1$, Definition 29 only involves $j = 1$. So e is good for $h - 1$ unless $r = h$, or $r \geqslant 3$ and $h - r$ is a perfect square. Note that $h - r$ is even. Suppose that $h - r = 4u^2$ with $u \geqslant 1$ and that $s \leqslant h - 3$. Then e lies in the rows 1 to $h - 3$, and for $g = h - 3$ the numbers $j(g - r + j)$ with $j = 1, 2, 3$ are

$$4u^2 - 2, \qquad 2(4u^2 - 1), \qquad 12u^2.$$

The first 2 are even and not divisible by 4, and the third is 3 times a square. None is a perfect square, so e is good for $h - 3$. This leaves the edges with $r = h$ and the edges

with $r = h - 4u^2 \geqslant 3$ and $s = h - 1$. If $r = h$, then $\rho(e)$ has the odd row 1 and is good for $h-1$. In the last case $\rho(e) = \{1+4u^2, 2\}$, where $1+4u^2 \leqslant h-2$. If $h-1-4u^2$ is not a perfect square, then $\rho(e)$ is good for $h-1$. Otherwise $h-(1+4u^2) = 4v^2$ with $v \geqslant 1$, and $\rho(e)$ is good for $h-3$ by the computation above. This proves the claim.

By the claim we can split the parity edges into 4 sets $\mathcal{E}_1, \ldots, \mathcal{E}_4$ such that the edges of $\mathcal{E}_1$ are good for $g_1 = h-1$, the edges of $\mathcal{E}_2$ are good for $g_2 = h-3$, and the edges $\rho(e)$ with $e \in \mathcal{E}_3$ or $e \in \mathcal{E}_4$ are good for $g_3 = h-1$ or $g_4 = h-3$, respectively. Put $N_k = \lceil c_0 \max(|\mathcal{E}_k|, 1)\, \tau^*(h)^2 \rceil$, and define

$$s_1 = 1, \qquad s_{k+1} = s_k + N_k + 8W_h, \qquad z = s_3 - 4W_h.$$

If $\mathcal{E}_k$ is not empty, apply Lemma 31 with $g = g_k$ and $N = N_k$, to $\mathcal{E}_k$ for $k = 1, 2$ and to $\rho(\mathcal{E}_k)$ for $k = 3, 4$. In both cases we get columns $X_e \in \{1, \ldots, N_k\}$ for $e \in \mathcal{E}_k$. Define a configuration B as follows.

- For $e \in \mathcal{E}_k$, column $x_e = s_k - 1 + X_e$ is black exactly in the 2 rows of e.
- For $e \in \mathcal{E}_k$ with $k \leqslant 2$ and $a \in A_e$, column $x_e + a$ is black exactly in row 1 if $a < 0$, and exactly in row g_k if $a > 0$.
- For $e \in \mathcal{E}_k$ with $k \geqslant 3$ and $a \in A_{\rho(e)}$, column $x_e + a$ is black exactly in row h if $a < 0$, and exactly in row $h + 1 - g_k$ if $a > 0$.
- Every other column x is black exactly in row 1 if $x < z$, and exactly in row h if $x \geqslant z$.

Put $U_k = [s_k - 2W_h,\, s_k + N_k - 1 + 2W_h]$ and $I_k = [s_k - 4W_h,\, s_k + N_k - 1 + 4W_h]$. By Lemma 30(a), the columns in the first 3 items for $\mathcal{E}_k$ lie in U_k. The intervals $I_1, \ldots, I_4$ are disjoint, the intervals I_1 and I_2 lie to the left of column z, and I_3 and I_4 start at column z or later. Hence B is well defined, and in the columns of I_k it satisfies the hypotheses of Lemma 32 for $k \leqslant 2$. For $k \geqslant 3$ the reflected configuration $\rho(B)$ satisfies them.

We show that B contains no obstruction Q. Recall that Q lies in $2W_h + 1$ consecutive columns. If Q has a point in a column of some U_k, then Q lies in the columns of I_k, which contradicts Lemma 32 for B or for $\rho(B)$. Otherwise every point of Q lies in row 1 and in a column $x < z$, or in row h and in a column $x \geqslant z$. These points form a chain, so Q is not a corner triple or a mixed quadruple by Lemma 27. It is not a rival pair by Lemma 25.

All doubles of B lie in the columns 1 to $M = s_4 + N_4 - 1$. For $m \geqslant M$ let S be the set of points of B in the columns 1 to m. As for even h, the blackout S is valid and has $m + q_h$ points. Finally

$$m_h \leqslant M = N_1 + N_2 + N_3 + N_4 + 24W_h \leqslant c_0(q_h + 4)\, \tau^*(h)^2 + 4 + 6h^2,$$

which is at most $C\, h^2\, \tau^*(h)^2$. This completes the proof of Theorem 2.

Remark 33. By a theorem of Wigert [11, Theorem 317], the bound of Theorem 2(b) is $h^2 \exp\bigl(O(\log h / \log\log h)\bigr)$. The factor $\tau^*(h)^2$ comes from worst cases, while the average of $\tau(n)$ over $n \leqslant x$ is about $\log x$. So it is natural to expect that averaged divisor sums, for example the bounds of Henriot [12], replace $\tau^*(h)^2$ by a power of $\log h$. We do not prove this here, and we do not know whether $m_h = O(h^2)$.

6 Small heights

We now prove Theorem 3. We describe a blackout by listing the black rows of its columns from left to right. For example, $14, 45, 25, 5$ has $B_1 = \{1, 4\}$ and $B_4 = \{5\}$.

6.1 Two, three, and four rows

Proof of Theorem 3(a). By Lemma 8, a tilted rectangle meets at least 3 rows. So on 2 rows every rectangle is axis-aligned. Let $m \geqslant 3$. A cycle in G_S needs 2 columns that are black in both rows. Hence, by Lemma 14, a blackout of the $2 \times m$ grid is valid if and only if at most 1 column is black in both rows. This gives $f(2, m) = m + 1$. For 3 rows, Theorem 15 gives $f(3, m) \leqslant m + 2$ and Lemma 20 gives $f(3, m) \geqslant m + 2$. □

On 4 rows, Lemma 8 gives $b + d \leqslant 3$. If $b = d = 1$, then $a = c = 1$, and the rectangle is a *diamond* $D(i, j)$ with corners $(i, j + 1)$, $(i + 1, j)$, $(i + 2, j + 1)$, and $(i + 1, j + 2)$, where $j \in \{1, 2\}$. Otherwise $\{b, d\} = \{a, c\} = \{1, 2\}$. This gives 4 rectangles A_i, B_i, C_i, and E_i with one corner in each of the columns i to $i + 3$, in the rows 3412, 3142, 2413, and 2143 from left to right.

Proposition 34. *For every $m \geqslant 6$ we have $f(4, m) = m + 4$.*

Proof. The upper bound is Theorem 15. For the lower bound take

$$S = \{(i, 1) : 2 \leqslant i \leqslant m - 1\} \cup \{(m - 1, 2), (m, 2), (1, 3), (2, 3), (1, 4), (m, 4)\}.$$

Then $|S| = m + 4$, and G_S is the cycle $r_1\, c_2\, r_3\, c_1\, r_4\, c_m\, r_2\, c_{m-1}$ of length 8 with the leaves $c_3, \ldots, c_{m-2}$ at r_1. So S is axis-valid by Lemma 14, and no axis-aligned rectangle is fully black.

A corner of a tilted rectangle in row 1 or 4 lies in a column from 2 to $m - 1$, so it is black in row 1 and white in row 4. Checking the other corners, we find that $D(i, 2)$, C_i, and E_i have at most 1 black corner, and that the tilted rectangles with 2 or more black corners are those in the following table. Here $s = 1$ for A_i and $s = 2$ for B_i.

rectangle	which i	white corners
$D(1, 1)$		$(1, 2), (3, 2)$
$D(i, 1)$	$i \in \{m - 3, m - 2\}$	$(i, 2), (i + 1, 3)$
A_i or B_i	$i \in \{1, 2\}$, $i \notin \{m - 4, m - 3\}$	$(i + s, 4), (i + 3, 2)$
A_i or B_i	$i \in \{m - 4, m - 3\}$, $i \notin \{1, 2\}$	$(i, 3), (i + s, 4)$
A_i or B_i	$i = 2$, $m = 6$	$(i + s, 4)$

The first 4 lines have different sets of rows. Within one line, the point in row 2 or 3 determines i, and the point in row 4 determines s. So these rectangles have distinct presentations.

It remains to show that no axis-aligned rectangle has one of these presentations. For $\{(1, 2), (3, 2)\}$ it would need a common black row of the columns 1 and 3, and there is none. In the next 3 lines the 2 white corners are opposite corners of one axis-aligned

rectangle, and one of its other corners is white. This corner is $(i, 3)$, $(i + s, 2)$, or $(i, 4)$, respectively, where we use $i \geqslant 3$, $i + s \leqslant 4 < m - 1$, and $3 \leqslant i \leqslant m - 3$. In the last line, the white corner is $(z, 4)$ with $z \in \{3, 4\}$. Column z is black only in row 1, and no column is black in the rows 1 and 4. So no axis-aligned rectangle shows only $(z, 4)$. □

Proof of Theorem 3(b). By Proposition 34 it remains to treat the widths 4 and 5. Lemma 20 gives $f(4, 4) \geqslant 7$ and $f(4, 5) \geqslant 8$. By Theorems 15 and 18, a valid blackout with $m+4$ points is one of $3\,(m)_4\,4^{m-4}$ candidates, that is one of 72 candidates for $m = 4$ and of 1440 for $m = 5$. A computer search finds 2 rectangles with the same presentation in each candidate. □

6.2 The finite collar theorem

We extend a finite blackout by columns that are black in one boundary row. For a blackout S of the $h \times m$ grid and $\lambda, \mu \geqslant 0$, let $S_{\lambda,\mu}$ be the blackout on the columns $1-\lambda, \ldots, m+\mu$ that agrees with S on the columns 1 to m and whose other columns are black exactly in a fixed *padding row*, which is row 1 or row h.

Theorem 35 (finite collar)**.** *Let $h \geqslant 3$ and let S be a blackout of the $h \times m$ grid. Then $S_{\lambda,\mu}$ is valid for all $\lambda, \mu \geqslant 0$ if and only if S_{W_h,W_h} is valid. Moreover $S_{0,\mu}$ is valid for all $\mu \geqslant 0$ if and only if S_{0,W_h} is valid, and likewise for $S_{\lambda,0}$.*

Proof. We prove the first statement. The others are proved in the same way. Put $S^* = S_{W_h,W_h}$, suppose that S^* is valid, and put $U = S_{\lambda,\mu}$. Call a tilted rectangle *weak* if it has 2 or more black corners. By (F1), a weak tilted rectangle of U has a black corner outside the padding row, hence in a column e with $1 \leqslant e \leqslant m$. Its span is at most W_h, so its columns lie between $1 - W_h$ and $m + W_h$. Hence it is also a rectangle of the grid of S^*, with the same presentation there, since U and S^* agree on their common columns.

Suppose that distinct rectangles R_1, R_2 have the same presentation P in U. By Lemma 6, $|P| \leqslant 2$.

Both axis-aligned. Every added column has degree 1 in G_U, so every cycle of G_U lies in G_{S^*}. Since S^* is valid and has 3 or more columns, Lemma 14 shows that G_{S^*} has no cycle of length 4 or 6, and then that U is axis-valid.

Both tilted. Both rectangles are weak, so both are rectangles of the grid of S^* with the presentation P, which contradicts the validity of S^*.

One of each. Let R_1 be tilted. It is weak, so it lies in the grid of S^*. A column of R_2 that contains a point of P is a column of R_1. A column of R_2 that contains no point of P has 2 black corners, so it is one of the columns 1 to m. Thus R_2 also lies in the grid of S^* and has the presentation P there, again a contradiction. □

6.3 Five rows

Let S_8 be the blackout $14, 45, 25, 5, 5, 34, 23, 12$ of the 5×8 grid, and let S_0 be the blackout $45, 34, 123, 1, 15$ of the 5×5 grid.

All validity claims below are finite checks from Definition 4, which we ran with several independent programs (Section 7).

Proof of Theorem 3(c). Let $m \geqslant 8$. The upper bound is Theorem 15. Since $W_5 = 5$, Theorem 35 with padding row 1 on the right reduces the lower bound to the validity of the test board $S_8, 1, 1, 1, 1, 1$ on the 5×13 grid. This board has 954 rectangles, and a direct check shows that their presentations are distinct.

For $m = 5$, Corollary 19 gives $f(5, 5) \leqslant 10$, and a direct check of its 130 rectangles shows that S_0 is valid. For $m = 6$ and $m = 7$ the blackouts $15, 12, 23, 3, 34, 45$ and $1, 12, 23, 3, 34, 45, 15$ are valid with $m + 5$ points, and checked SAT certificates exclude $m + 6$ points (Section 7). □

Thus $f(5, 8) = 14$ is the first value with $f(h, m) > h+m$ among the grids with $h \leqslant m$, in the order of increasing area.

6.4 Six and seven rows

Let S_{16} be the blackout $36, 14, 25, 6, 6, 16, 6, 6, 56, 45, 4, 4, 4, 34, 23, 12$ of the 6×16 grid. Its 9 doubles carry the edges of the complete bipartite graph between $\{1, 3, 5\}$ and $\{2, 4, 6\}$. To exclude blackouts with 1 point less than the row bound, we first find their row graphs.

Lemma 36. *Let $m \geqslant 3$, and let S be an axis-valid blackout of the $6 \times m$ grid with $m + 8$ points. Then every column of S is empty, a singleton, or a double, the doubles have distinct row pairs, and one of the following holds.*

(a) No column is empty, and the pairs of the doubles are the edges of $K_{3,3}$ minus an edge.

(b) No column is empty, and the pairs of the doubles are the edges of $K_{2,4}$.

(c) Exactly 1 column is empty, and the pairs of the doubles are the edges of $K_{3,3}$.

Proof. For every choice of the trees, Lemma 16 gives $|S| = t + e(H)$ with $t \leqslant m$, and $e(H) \leqslant 9$ by Lemma 17. So $(t, e(H))$ is $(m, 8)$ or $(m-1, 9)$. In the second case, the proof of Theorem 18 shows that H is a $K_{3,3}$ and that no column has 3 black rows, which gives (c).

In the first case H is triangle-free with 8 edges. If it were not bipartite, its shortest odd cycle would be a cycle of length 5 without chords. The sixth vertex is joined to at most 2 vertices of this cycle, so $e(H) \leqslant 7$, a contradiction. Hence H is bipartite with parts of sizes k and $6 - k$, and $k(6 - k) \geqslant 8$ shows that H is $K_{2,4}$ or $K_{3,3}$ minus an edge. In both graphs, 2 vertices of the same part have 2 or more common neighbors. Suppose that $|B_i| \geqslant 3$. Then T_i has a vertex v with 2 neighbors u and w. These lie in one part, so they have a common neighbor $z \neq v$. Replacing vw by uw in T_i gives another tree on B_i and the triangle uwz in H, which contradicts Lemma 16. Thus every column is a singleton or a double, and the pairs of the doubles are the edges of H. □

There are 90 labeled graphs in case (a), 15 in case (b), and 10 in case (c).

Proof of Theorem 3(d). Lower bounds. The blackouts $14, 1, 12, 2, 23, 3456$ of the 6×6 grid, $16, 1, 1, 12, 2, 23, 3456$ of the 6×7 grid, and $15, 1, 123, 34, 4, 4, 456, 26$ of the 6×8 grid are valid with 12, 13, and 15 points. The blackout $12, 23, 34, 45, 6, 6, 36, 56, 25$ of the 6×9 grid has 16 points. Its test board with $W_6 = 7$ columns black only in row 1 on each side is valid, so Theorem 35 gives $f(6, m) \geqslant m+7$ for every $m \geqslant 9$. For $12 \leqslant m \leqslant 14$ the blackout $16, 36, 25, 12, 1^{m-8}, 23, 34, 45, 56$ is valid with $m + 8$ points, where 1^k stands for k columns black only in row 1. For $m = 15$ the blackout $56, 45, 34, 23, 1, 1, 14, 1, 1, 12, 36, 1, 25, 6, 16$ is valid with 24 points. For $m \geqslant 16$ we apply Theorem 35 to S_{16} with padding row 1 on the right. Its test board on the 6×23 grid has 4627 rectangles with distinct presentations.

Upper bounds. Theorem 15 gives $f(6, m) \leqslant m + 9$. For $m = 6, 7, 8$, checked SAT certificates exclude 13, 14, and 16 points. Let $9 \leqslant m \leqslant 14$. A valid blackout with $m + 9$ points would have doubles on the edges of a $K_{3,3}$ by Theorem 18. Checked SAT certificates exclude each of the 10 choices of the $K_{3,3}$ (Section 7). Hence $f(6, m) \leqslant m + 8$. Let $9 \leqslant m \leqslant 11$. By Lemma 36, a valid blackout with $m + 8$ points has one of 115 row graphs. Checked SAT certificates exclude each of them, so $f(6, m) \leqslant m + 7$. □

Proof of Theorem 3(e). The blackouts $17, 1, 12, 2, 23, 3456, 67$ of the 7×7 grid and $27, 367, 46, 2, 6, 24, 125, 16$ of the 7×8 grid are valid with 14 and 16 points, and checked SAT certificates exclude 15 and 17 points. Let $m \geqslant 24$. The row bound gives $f(7, m) \leqslant m + 12$. The file `papers/extremal/data/seven_row_blackouts.txt` of the repository [24] lists 5 blackouts of the widths 24 to 28 that are valid with $m + 12$ points. For example, the one of width 24 is

$$56, 45, 34, 23, 1, 1, 14, 1, 12, 1, 6, 16, 7, 1, 47, 6, 67, 7, 7, 6, 25, 7, 36, 27.$$

The one of width 28 stays valid after adding $W_7 = 10$ columns black only in row 7 on its left. Hence Theorem 35, with padding row 7 on the left, gives $m + 12$ points for every $m \geqslant 28$. □

7 Computer checks

The computer-assisted parts of Theorem 3 are of 2 kinds. The lower bounds need the validity of explicit blackouts, including the test boards of Theorem 35. The upper bounds below the row bound need the unsatisfiability of SAT instances. The repository [24] contains the code, the evidence files, and the commands. The scripts are in `jis/scripts`, the evidence in `jis/data`, the certificate pipeline in `jis/unsat`, and the formal proofs in `lean/KageyProblems/Problem001`.

Validity of explicit blackouts. Every blackout and test board of Section 6 was checked from Definition 4 by at least 2 separately written programs, which list all rectangles and compare their presentations. The script `papers/extremal/scripts/check_constructions.py` is one more such program. It checks all blackouts of Section 6, including the 5 blackouts on 7 rows in `papers/extremal/data/seven_row_blackouts.txt`, the blackouts of Proposition 21 for $3 \leqslant h \leqslant 8$, and blackouts built as in Section 5 for

$4 \leqslant h \leqslant 13$. In the last test the columns X_e are found by a random search in a shorter window, and the singletons in (3) of Lemma 32 are chosen at random. For $h = 5$ and $h = 7$, where $g = h - 3$ is not allowed, the sea rows $\{1, h-1\}$ and $\{2, h\}$ suffice, and the test uses the same layout with 2 blocks. The 72 and 1440 candidates on the 4×4 and 4×5 grids are searched by `verify_narrow.py`.

SAT encodings. Encoding A has a variable for each grid point and a clause for each inclusion-minimal set of the form $C(R_1) \triangle C(R_2)$ with $R_1 \neq R_2$. The clause says that this set is not fully black. Encoding B has such clauses only for the pairs of rectangles that share a corner. It also has a variable y_R for each rectangle R, which is forced to be true if R is fully black, and it allows at most 1 true y_R. A *structured* instance on 6 rows fixes one row graph from Theorem 18 or Lemma 36. Each column chooses a type, which is a single row, an edge of the graph, or the empty set in case (c) of the lemma. Each edge is used once, and the empty type is used once in case (c). The instance keeps those clauses of encoding B that involve a tilted rectangle and whose set can be fully black for these types. The size of the blackout is fixed by a cardinality constraint.

These encodings are faithful for the following reasons. By Lemma 5, a blackout is valid if and only if it contains no set $C(R_1) \triangle C(R_2)$ with $R_1 \neq R_2$, and it is enough to exclude the inclusion-minimal sets. This is encoding A. For rectangles without a common corner, the set $C(R_1) \triangle C(R_2)$ is black exactly when both are fully black. Hence a blackout is valid if and only if at most 1 rectangle is fully black and no set $C(R_1) \triangle C(R_2)$ of 2 rectangles with a common corner is black. This is encoding B. Finally let S be a valid blackout of the $6 \times m$ grid with $m + 9$ or $m + 8$ points, where $m \geqslant 3$. Theorem 18 or Lemma 36 gives its row graph, and every B_i is an allowed type. With y_R true exactly for the fully black rectangles, S satisfies encoding B, and the structured instance of its row graph has a subset of these clauses. So if all structured instances of a width and a size are unsatisfiable, there is no valid blackout of that width and size.

Certificates. CaDiCaL [3] produced LRAT proofs for every unsatisfiable instance that we use. These are the instances of encoding A for $f(5, 6) < 12$, $f(5, 7) < 13$, $f(6, 6) < 13$, $f(6, 7) < 14$, $f(7, 7) < 15$, $f(6, 8) < 16$, and $f(7, 8) < 17$, the 70 structured instances with $m+9$ points for the widths 8 to 14, and the 345 structured instances with $m+8$ points for the widths 9 to 11. The instances for 6×8, 7×7, and 7×8 are split by the content of the first column. Each of the 739 proofs was accepted by 2 checkers, the program `lrat-check` of drat-trim [5, 28] and the LRAT checker of the Lean core library. The proofs are large and are not stored, but their hashes are, and the pipeline recreates them.

Formal proofs. The Lean development contains formal proofs of Lemmas 5, 6, and 9, of the row bound, of the finite collar theorem, and of Proposition 21 under the stronger assumption $p_{j+1} - p_j \geqslant 2W_h + 2$. This gives a formal proof of $f(h, m) = m + q_h$ for every $h \geqslant 3$ and $m \geqslant q_h + (q_h - 1)(2W_h + 1)$. The spacing $W_h + 1$ of Proposition 21 and the results of Section 5 are not formalized. Validity tests of S_0, S_8, the 6×8 and 6×15 blackouts of Section 6, and the 5×13 and 6×23 test boards are checked by the Lean kernel. Together with the formal row bound and collar theorem, this proves $f(5, m) = m + 6$ for $m \geqslant 8$,

$f(6, m) = m + 9$ for $m \geqslant 15$, $f(5, 5) \geqslant 10$, and $f(6, 8) \geqslant 15$ in Lean.

8 Square grids

On the $n \times n$ grid the row bound is of order n^2 and is far from the truth. For $n \geqslant 3$, Lemma 14 and Neuwirth's bound for bipartite graphs of girth 8 with n vertices on each side [21, Corollary 16] give $f(n, n) < n^{4/3} + \frac{2}{3}n$. The incidence graphs of generalized quadrangles show that this is the right order for axis-aligned rectangles alone [2]. With tilted rectangles we only have the following lower bound.

Remark 37. There is a constant $c > 0$ such that $f(n, n) \geqslant c\, n^{6/5}(1 + \ln n)^{-2/5}$ for every $n \geqslant 2$. In particular $f(n, n) - 2n \to \infty$. We only sketch the proof, which is the deletion method [1, Section 3.2]. Put $\Lambda = 1 + \ln n$. By Lemmas 5, 6, and 7, a blackout is valid if it contains no set of the following 2 families. The family $\mathcal{A}$ consists of the corner sets of rectangles and of the supports of mixed pairs, which have 4 points. The family $\mathcal{B}$ consists of the sets $C(R_1) \triangle C(R_2)$ for rectangles with exactly 1 common corner, which have 6 points. A count of the sides at a point shows that at most $8n^2\Lambda$ rectangles have a corner at a given point. This gives $|\mathcal{B}| \leqslant 32n^6\Lambda^2$, and with the divisor bound it gives $|\mathcal{A}| \leqslant Kn^{4+1/10}\Lambda$. Choose each point independently with probability $a\, n^{-4/5}\Lambda^{-2/5}$, where $a = 192^{-1/5}$, and delete 1 point from each member of $\mathcal{A} \cup \mathcal{B}$ whose points are all chosen. The expected number of points that remain is at least $(\frac{5}{6}a - o(1))\, n^{6/5}\Lambda^{-2/5}$. The exponent $6/5$ comes from the roughly n^6 members of $\mathcal{B}$.

9 Open problems

(1) *The order of the threshold.* Theorem 2 gives $q_h \leqslant m_h \leqslant h^{2+o(1)}$. Is $m_h = O(h^2)$? The known values are $m_4 = 6$, $m_5 = 8$, and $m_6 = 15$, and we know that $m_7 \leqslant 24$. What is m_7?

(2) *A lower bound.* The bound $m_h \geqslant q_h$ only counts the doubles. Is there a constant $c > 1$ with $m_h \geqslant c\, q_h$ for all large h? Does m_h/q_h tend to infinity?

(3) *Monotonicity.* Is $f(h, m + 1) \geqslant f(h, m) + 1$ for all $h \geqslant 2$ and $m \geqslant 3$? This holds for all values in Theorem 3. It would imply that the widths with $f(h, m) = m + q_h$ are exactly the widths $m \geqslant m_h$. We could not even prove that $f(h, m)$ is nondecreasing in m. The difficulty is that a new column creates new tilted rectangles with 3 corners in the old columns.

(4) *Square grids.* What is the order of growth of $f(n, n)$, between $n^{6/5-o(1)}$ and $n^{4/3}$? Theorem 3 gives $f(n, n) \leqslant 2n$ for $n \leqslant 7$. What is the first n with $f(n, n) > 2n$?

(5) *Seven rows.* What is $f(7, m)$ for $9 \leqslant m \leqslant 23$?

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